

# Wasserstein Kmeans Algorithm

Implementing clustering algorithm over probability density functions

## Mathematical Overview

The purpose of this algorithm is to find an optimal way to cluster  $N$  PDFs (probability density functions) using the second Wasserstein distance function, defined formally as:

$$W_2^2(P, Q) = \inf_{\gamma \in \Gamma(P, Q)} \int_{\mathbb{R} \times \mathbb{R}} |x - y|^2 d\gamma(x, y)$$

where  $P$  and  $Q$  are PDFs, and  $\Gamma(P, Q)$  is the set of joint distributions with marginal density functions  $P$  and  $Q$  (Le Gouic and Loubes 2016).

We know from [reference] that if  $P$  and  $Q$  are univariate:

$$W_2^2(P, Q) = \int_0^1 (F_P^{-1}(u) - F_Q^{-1}(u))^2 du$$

where  $F_X^{-1}(u)$  is the inverse CDF (cumulative distribution function) for PDF  $X$ .

The task of finding a centroid for  $N$  data points can be achieved by finding  $Y$  such that:

$$Y = \arg \min_{Y \in B} \sum_{i=1}^N d(X_i, Y)$$

where our data points exist in set  $S \subseteq B$ , and  $d$  is an arbitrary distance function.

Let  $d = W_2^2$  (the squared Wasserstein distance). Then we wish to find a point  $Y$  such that:

$$Y = \arg \min_{Y \in B} \sum_{i=1}^N \int_0^1 (F_Y^{-1}(u) - F_{X_i}^{-1}(u))^2 dt$$

From [reference] we know that:

$$F_Y^{-1} = \frac{1}{N} \sum_{i=1}^N F_{X_i}^{-1}(u)$$

Thus, we can easily find a the best fit cluster of our PDFs by averaging the quantile functions of our data points. If we wanted to assign each point to a set of  $K$  clusters, the objective is to find a sequence of  $k$  clusters  $\mu_1, \dots, \mu_K$ :

$$\mu_1, \dots, \mu_K = \arg \min_{\mu_1, \dots, \mu_K} \sum_{k=1}^K \sum_{X_i \in C_k} d(X_i, \mu_k)$$

We can find a local minimum to this problem if, at step  $t + 1$ , we update each cluster using update procedure:

$$\mu_k^{t+1} \leftarrow \frac{1}{|C_k^t|} \sum_{X_i \in C_k^t} F_{X_i}^{-1}(u)$$

## Sources

Le Gouic, Thibaut, and Jean-Michel Loubes. 2016. “Existence and Consistency of Wasserstein Barycenters.” *arXiv Preprint arXiv:1506.04153*. <https://arxiv.org/abs/1506.04153>.